

StudyMate AI: Operating Systems and Process Scheduling

Chapter 1: Process Management and CPU Scheduling

An Operating System kernel manages the execution of active processes.

A process transitions through states: Ready, Running, and Blocked.

Core CPU Scheduling Algorithms:

1. First-Come, First-Served (FCFS): Non-preemptive. Subject to the Convoy Effect.
2. Shortest Job First (SJF): Optimal for minimizing average wait time.
3. Round Robin (RR): Preemptive scheduling with a fixed Time Quantum (10-100ms).
4. Multilevel Feedback Queue (MLFQ): Adjusts priorities based on CPU burst history.

Key Metrics:

Turnaround Time: Submission to completion time.

Waiting Time: Total time spent in ready queue.

Throughput: Processes completed per unit of time.

Chapter 2: Google File System (GFS) Architecture

Google File System is a distributed file system designed for large data clusters.

Key Architectural Principles:

1. Commodity Hardware: Component failures are treated as the norm.
2. Large Files: Optimized for files of gigabytes to terabytes in size.
3. Append-Only Mutations: Multiple clients append concurrently.
4. Large Chunk Size: Files are divided into 64 MB chunks.

GFS Components:

- Master Node: Stores all metadata in memory and coordinates chunkservers.
- Chunkservers: Store 64MB chunks as Linux files with 3x replication.
- Client: Communicates with Master for metadata, reads directly from Chunkservers.